Small-variable polynomial constructions over the rationals

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Abstract

We construct integer polynomials in 29 and 31 total variables by eliminating finite algebraic choices from an explicit arithmetic system. Their degrees are bounded by $2^{2093605}$ and 2^{180}, respectively. We prove the polynomial constructions, their numerical bounds, the local input normalization and an exact index-depth formula. Under four precisely stated arithmetic inputs their rational input projections equal the integers. Every finite expression is specified by reproducible data, with independent algebraic checks.

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1 The constructions and their scope

For a polynomial $P \in \mathbb{Z}[T, Y_1, \ldots, Y_m]$, its rational input projection is the set of $t \in \mathbb{Q}$ for which some rational tuple $\mathbf{y}$ satisfies $P(t, \mathbf{y}) = 0$. We count T among the variables. The degree is the total degree, and the coefficient height is the largest absolute value of an integer coefficient.

We give two complete finite specifications with 29 and 31 total variables. Their polynomial character and numerical bounds are proved by finite algebra. The assertion that their input projections equal $\mathbb{Z}$ is a corollary of the four arithmetic inputs in Section 4. Those inputs are stated in full and attributed to their sources. They are not inferred from the coefficient checks.

The construction separates retained rational coordinates from finite algebraic choices. Monic equations for the latter define a finite free algebra. A separating linear form and a reciprocal equation then replace all its finite choices by one additional rational coordinate. The resulting formulas are actual integer polynomials: their determinants and finite products do not introduce quantified roots.

2 The exact numerical height constant

The contact-height input uses a fixed, explicitly specified integer. Its exact evaluation improves the numerical bounds in both construction methods. Put

$$B_* = \prod_{p \le 76176} p, \qquad H_{19} = 2^{110} 210^{1800} B_*^{19} + 128.$$

This is the constant of the source's exponent-nineteen height theorem [1, Theorem `thm: quartic-height19`].

Lemma 2.1. *The exact inequalities*

$$2^{2093447} < H_{19} < 2^{2093448}$$

hold. In particular one may use $h = 2093448$ in every argument requiring $H_{19} < 2^h$.

Proof. This finite integer calculation has the following reproducible certificate. Enumerate primes through 76176 by marking, for each unmarked $p \le \lfloor \sqrt{76176} \rfloor$, all multiples starting at p^2. The remaining integers at least two are exactly the primes: every composite has a prime divisor not exceeding its square root, and no prime is marked. An independent enumeration tests every integer by trial division through its integer square root. Both lists have 7499 entries and agree entry by entry.

Their exact product has 109445 binary digits. Computing the displayed powers and product as integers gives 2093448 binary digits for H_{19}. The comparison with the next smaller power of two is strict as well. The supplied checker performs both prime enumerations and both integer comparisons; the complete prime list and the unsigned binary encoding of H_{19} are included. No floating-point logarithm enters these comparisons. □

This lemma evaluates a constant. It does not prove the contact-height inequality itself. The latter is still the arithmetic input stated in Section 4, now used with this exact value and its sharper certified bound.

3 An explicit polynomial for a guarded finite algebra

Let $R = \mathbb{Z}[x_1, \ldots, x_b]$. Consider a triangular monic presentation

$$A = R[z_1, \ldots, z_m]/(f_1, \ldots, f_m), \qquad f_i \in R[z_1, \ldots, z_i],$$

where f_i is monic of degree $e_i \geq 1$ in z_i. Its standard monomials $\prod_i z_i^{a_i}$, $0 \leq a_i < e_i$, form an R-basis. In particular A is free of rank $n = \prod_i e_i$. For $a \in A$, write M_a for its multiplication matrix in this ordered monomial basis. The matrices M_a commute, and their entries are integer polynomials in the base coordinates. They are completely specified by repeated reduction using the monic equations; they need not be materialized to specify a determinant involving them.

Fix elements $G, E_1, \ldots, E_r \in A$. We assume that in every geometric fiber, every point at which G is nonzero has local length one. This condition holds, for example, if G contains the product of the diagonal derivatives $\partial f_i/\partial z_i$ as a factor. We seek the rational base points admitting a rational point of A with

$$G \neq 0, \qquad E_1 = \cdots = E_r = 0.$$

For $0 \leq \ell \leq mn(n+1)/2$, define

$$\lambda_\ell = 1 + \sum_{i=1}^{m} \ell^i z_i, \qquad M = M_{G\lambda_\ell}, \qquad H(V) = \det(I - VM) = \sum_{k=0}^{n} a_k V^k.$$

Here and below $a_k \in R$ depend on the fixed integer ℓ; always $a_0 = 1$. Also put

$$R_i(V) = \operatorname{Tr}\big(M_{E_i} \operatorname{adj}(I - VM)\big).$$

This is a polynomial in $R[V]$ of degree at most $n-1$.

For each $1 \leq d \leq n$, set

$$\begin{aligned}
H_d(V) &= \sum_{k=0}^{d} a_k V^k, \\
g &= a_d \operatorname{Res}_V(H_d, H_d'), \qquad L = a_d^2 g, \\
\Psi(U) &= 1 + \sum_{j=1}^{d} a_{d-j} a_d^{2j-1} g^j U^j, \\
\widetilde{R}_i(U) &= (LU)^{n-1} R_i\big(1/(LU)\big), \\
Q_{\ell,d}(\mathbf{x}, U) &= \Psi(U)^2 + \sum_{k=d+1}^{n} a_k^2 + \sum_{i=1}^{r} \widetilde{R}_i(U)^2.
\end{aligned}$$

The resultant is the Sylvester determinant with the specified degree bounds d and $d-1$. The expression for $\widetilde{R}_i$ denotes polynomial coefficient reversal, hence defines a polynomial also at $LU = 0$. Every displayed expression has integer coefficients.

Theorem 3.1. *The explicitly specified polynomial*

$$P(\mathbf{x}, U) = \prod_{\ell=0}^{mn(n+1)/2} \prod_{d=1}^{n} Q_{\ell,d}(\mathbf{x}, U)$$

has the following property. For every $\mathbf{a} \in \mathbb{Q}^b$, there exists $U \in \mathbb{Q}$ with $P(\mathbf{a}, U) = 0$ if and only if there is a rational point of the fiber algebra $A_{\mathbf{a}}$ satisfying $G \neq 0$ and all $E_i = 0$. The formula uses only one coordinate in addition to the b base coordinates; its formal algebra generators are eliminated by the determinants and are not quantified coordinates.

Proof. Fix a rational base point. First consider one factor $Q_{\ell,d}$. At $a_d g = 0$ or $U = 0$, its reciprocal residual Ψ equals one. Otherwise, with $V = 1/(LU)$,

$$\Psi(U) = (LU)^d \frac{H_d(V)}{a_d}.$$

Thus a rational zero of $Q_{\ell,d}$ implies $a_d g U \neq 0$, all coefficients a_k with $k > d$ vanish, and $H(V) = H_d(V) = 0$. The resultant condition makes this a simple root of H. Since $H(0) = 1$, the root V is nonzero.

The rational matrix $I - VM$ has a one-dimensional kernel. Every multiplication matrix M_{z_i} preserves it and therefore acts on it by a rational scalar ζ_i. Applying the algebra relations to this common eigenline shows that $(\zeta_1, \ldots, \zeta_m)$ is a rational point of the fiber. Its $G\lambda_\ell$-value is $1/V$, so its G-value is nonzero.

To verify the closed conditions, write $J = \mathrm{Tr}(\mathrm{adj}(I - VM))$. At a root of H, the adjugate identity gives $M \,\mathrm{adj}(I - VM) = V^{-1} \,\mathrm{adj}(I - VM)$. Differentiating the determinant then yields

$$J = -VH'(V) \neq 0.$$

The columns of the adjugate lie in the common eigenline, so

$$R_i(V) = E_i(\boldsymbol{\zeta}) J.$$

Since $LU \neq 0$, the equation $\widetilde{R}_i(U) = 0$ is equivalent to $R_i(V) = 0$, and hence $E_i(\boldsymbol{\zeta}) = 0$. This proves the reverse implication without assuming that the rest of the fiber algebra is reduced.

Conversely, suppose a suitable rational point exists. Over an algebraic closure, multiplication by $G\lambda_\ell$ has the value $G(p)\lambda_\ell(p)$ at every geometric point p, with multiplicity equal to its local length. If $G(p) = 0$, its factor in $\det(I - VM)$ is one, even if the local algebra is nonreduced. If $G(p) \neq 0$, its local length is one by hypothesis.

At any such point p, the polynomial $G(p)(1 + \sum_{i=1}^{m} \ell^i z_i(p))$ is nonzero as a polynomial in ℓ, because its constant coefficient is $G(p) \neq 0$. For distinct such points p, q, their difference is also nonzero: if all coefficients vanished, first $G(p) = G(q) \neq 0$, and then all their coordinates would agree. Each of these polynomials has degree at most m. With at most n geometric points, there are at most $mn(n+1)/2$ bad values of ℓ in total, counting zero values and collisions. Thus one of the specified integers makes all nonzero guarded eigenvalues distinct and nonzero.

For that ℓ, the polynomial H is squarefree of some degree $1 \leq d \leq n$. The given rational point provides a nonzero rational root V of H. Its closed residuals vanish, so the trace identities give $R_i(V) = 0$. The leading coefficient and resultant are nonzero. Taking $U = 1/(LV)$ therefore gives a zero of the corresponding factor $Q_{\ell,d}$, and hence of P. □

The degree cases are part of the literal finite formula. They include every exceptional drop in degree of H. The discriminant condition excludes bad separating choices, and the finite bound on ℓ proves that this exclusion loses no valid rational point. In particular the construction does not leave a generic primitive-element argument or an unperformed exceptional-stratum decomposition implicit.

4 Odd products and a 29-variable construction

Variable counts include the distinguished input. The construction uses Conant's odd-product height identity and direct-target comparison argument as presented in [1, Lemma `v9:lem:product` and Lemma `lem:v9-bridge`], together with the explicit finite algebra construction. We give the rational system, prove its input projection under precisely stated arithmetic hypotheses, and then eliminate its finite algebraic choices. The product identity and point charts are proved independently of the contact-height and prime-pattern inputs.

4.1 Arithmetic inputs and conventions

Put

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and fix

$$R_0 = (1+\sqrt{2}, 2\sqrt{2}), \qquad P = 2R_0 = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

Write σ for conjugation, $T_0 = (0,0)$, $f(Q) = 1/x(Q)$, and $z(Q) = -x(Q)/y(Q)$. Heights are absolute logarithmic Weil heights, with normalized local degree weights. Valuations at unramified primes satisfy $v(p) = 1$.

We use the following arithmetic hypotheses, with the named source locations in [1].

1. (Elliptic group and height.)
   Source: `subsec:v5-point-source`, `eq:o8-indexheight`.

   $2E(K) = \mathbb{Z}P$, and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.

2. (Contact height: Theorem `thm:quartic-height19`.) For $t = u/v$ in lowest terms, $v > 0$, and $t \notin \mathcal{D} = \{0, 1, 2401, -343/32\}$, define $M_5(t)$ to be the product of the primes $p > 76176$ for which at least one of

   $$v_p(v), \quad v_p(u), \quad v_p(u-v), \quad v_p(u-2401v), \quad v_p(32u+343v)$$

   is positive and odd. These are the normalized contacts with $\infty, 0, 1, 2401, -343/32$. Put

   $$B_* = \prod_{p \leq 76176} p, \qquad H = H_{19} = 2^{110} 210^{1800} B_*^{19} + 128.$$

   The inherited assertion is $h(t) \leq H M_5(t)^{19}$. Exact integer evaluation gives $H < 2^{2093448}$. In either profile below we may use the indicated weaker exponent η.

3. (Pole obstruction and realization.)
   Source: `hyp:o8-pole` and `lem:original-seven-simultaneous-realization`. The pole equations, with $B = 17b$,

   $$U = b\frac{1-bt^2}{1+bt^2}, \quad h_0 = e(B-U^2), \quad p^2 = B(1-e)(1+3e), \quad Bq^2 = (s^2 - Bh_0)(s^2+3Bh_0),$$

   exclude negative odd $v_\ell(b)$ for $\ell \notin \{2, 3, 17\}$. They are soluble with $t = 0$ whenever b has square reduced denominator, is positive and less than 17, is square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and satisfies $v_2(b) \geq 3$, $v_3(b) \geq 1$, $v_{17}(b) \geq 2$.

4. (Simultaneous prime patterns: Theorem `thm:v5-simultaneous-primes`, Lemma `lem:v5-two-reductions`, and the application of [2, Theorem 4.1].) The simultaneous Gaussian prime-pattern input supplies the prescribed prime values $u + c_j w^2$, arbitrary fixed divisibility of w, and arbitrarily large u/w^2. Their two conjugate reduction orders satisfy

   $$v_2(d_-) = 6, \quad v_2(d_+) = 2, \quad \gcd(d_-(p), d_+(p')) = 4$$

   for every pair of selected primes. An arbitrary fixed odd avoidance modulus can be imposed. For each selected Gaussian prime π, the source convention is $\pi \equiv 1+3i \pmod 7$, making both $N(\pi - 1)$ and $N(\pi + 1)$ units at seven. At each other odd prime ℓ in the fixed avoidance modulus, the condition $\pi \equiv i \pmod \ell$ makes both orders congruent to two modulo ℓ. Thus the orders avoid the entire odd part of the fixed modulus. The selected rational primes are 1 modulo $8 \cdot 3 \cdot 17$.

These hypotheses include the simultaneous application of Sun's stated prime-pattern theorem. They are assumptions of the theorem below, not conclusions of its implementation audit. The semilocal index-size condition is a separate elementary consequence of the displayed curve and point, proved in Proposition 8.1; it is not an additional arithmetic hypothesis.

The two final profiles are fixed as follows. The symbols μ, κ are assigned divisibility orders, while ρ_0, ρ_1 are actual formal logarithm truncation precisions. The scalar ratio order is m.

variables	η	μ	κ	ρ_0	ρ_1	m	size certificate
29	24	24	128	24	128	100485552	$M = 2^{2093449}$
31	19	10	39	12	40	20	$C = 3^{2^{26}}$.

For the first profile the height exponent 24 is a consequence of the exponent-19 input. No new arithmetic hypothesis is introduced by the large scalar order. Section 6 proves the size and precision inequalities for both profiles.

4.2 An elementary Lucas growth estimate

The following estimate is independent of the arithmetic hypotheses. It supplies the growth inequality used in the integer-extraction argument.

Lemma 4.1. *Let $a, t, b \in \mathbb{Z}$, with $|a| > 7$ and $b \neq 0$, satisfy*

$$t^2 - a(a-2)tb + (a-2)^2b^2 = 1.$$

Then $\log|t| > |a| - 3$. In particular, $\log\max(|a|, |t|, |b|) > |a| - 3$.

Proof. Put $A = |a|$, $c = (a-2)b$ and $c' = \operatorname{sgn}(a)c$. Then

$$t^2 - Atc' + (c')^2 = 1.$$

We have $|c'| \geq |a-2| \geq 6$, so $t \neq 0$. The two terms t, c' have the same sign: opposite signs would make the left-hand side at least $t^2 + (c')^2 > 1$. Thus $T = |t|$ and $C = |c'|$ are positive integer solutions of $T^2 - ATC + C^2 = 1$.

Define $U_0 = 0$, $U_1 = 1$ and $U_{j+1} = AU_j - U_{j-1}$. Every positive solution consists of adjacent terms of this sequence. Indeed, order its coordinates as $M \geq m > 0$. The integer

$$Am - M = \frac{m^2 - 1}{M}$$

lies in $[0, m)$, and replacing (M, m) by $(m, Am - M)$ preserves the equation. Repeated descent ends at $(1, 0)$. Reversing the steps gives the unordered pairs $\{U_{j+1}, U_j\}$. Consequently $C = U_j$ for some $j \geq 1$ and $T \geq U_{j-1}$.

Put $d = |a-2|$. Since $d \mid C$, reduction of the recurrence modulo d gives $d \mid j$: if $a > 0$, then $A \equiv 2 \pmod{d}$ and $U_j \equiv j \pmod{d}$; if $a < 0$, then $A \equiv -2 \pmod{d}$ and $U_j \equiv (-1)^{j-1}j \pmod{d}$. Hence $j \geq d \geq A - 2$. The recurrence also gives $U_r > (A-1)^{r-1}$ for $r \geq 2$. This follows by induction from $U_2 = A > A - 1$ and $U_{r+1} > (A-1)U_r$, since $U_r > U_{r-1}$. As $j \geq 6$, we obtain

$$\log T > (j-2)\log(A-1) \geq (A-4)\log 7 > A - 3 \qquad (A \geq 8).$$

The last inequality holds at $A = 8$ and its difference increases with A. This proves the assertion. □

4.3 An exact height identity

Lemma 4.2. *For any nonzero odd integers $n_1, \ldots, n_r$,*

$$h\left(\prod_{i=1}^{r} x([n_i]P)\right) = \sum_{i=1}^{r} h(x([n_i]P)).$$

This identity requires only that the displayed $P = 2R_0$ have infinite order; it does not require the contact-height or prime-pattern hypotheses.

Proof. First consider a finite place v with $v(D) = 0$. The formal subgroup $E_1(K_v)$ consists of the identity and points with $v(x) < 0$. It is an open subgroup, so $E(K_v)/E_1(K_v)$ is finite. These facts hold for the displayed integral Weierstrass equation, including the place above two. The image of T_0 has order two, since $T_0 \notin E_1(K_v)$, and

$$x(Q + T_0) = -D/x(Q).$$

If an odd multiple of P belongs to $E_1(K_v)$, the image of P in the quotient has odd order. No multiple of that image is the nonzero element represented by T_0. Hence no odd multiple of P has positive x-valuation. If no odd multiple belongs to $E_1(K_v)$, all odd multiples have nonnegative x-valuation. Thus the numbers $v(x([n_i]P))$ have a common weak sign.

The only remaining finite place is the prime $(3 - \sqrt{2})$ above seven. Here $v(D) = 2$, and the discriminant has valuation six, so the displayed integral model is minimal. Its reduction is $y^2 = x^3$. The displayed P reduces to $(1, 6)$, a nonsingular point. The subgroup of points with nonsingular reduction therefore contains every multiple of P. A point with positive x-valuation would have positive y-valuation and reduce to the singular point $(0, 0)$. Consequently every multiple of P has x-valuation at most zero at this place.

At either real embedding every $[n_i]P$ is a nonzero double. The duplication identity

$$x(2Q) = \left(\frac{x(Q)^2 + D}{2y(Q)} \right)^2$$

makes its abscissa positive. The curve equation and its nonzero ordinate imply $x([n_i]P) > d > 1$ in each embedding. Each real local height therefore adds under multiplication. At a finite place the common weak sign similarly gives

$$\max\{0, -\textstyle\sum_i v(x([n_i]P))\} = \sum_i \max\{0, -v(x([n_i]P))\}.$$

Summing with the normalized local weights proves the identity. □

Corollary 4.3. *Let $Q_1, \dots, Q_4$ be odd multiples of P and write*

$$C = \prod_{i=1}^{4} x(Q_i) = C_0 + C_1\sqrt{2}.$$

The three numbers $A_0 = 1$, $A_1 = C_0^2$, and $A_2 = (1 + C_1^2)^2$ are nonzero positive rational squares. If the three contact detectors at $A_j\lambda$ imply $M_0(A_j\lambda)^k \mid N \neq 0$, then

$$h(x(Q_i)) < 7H|N|^{\eta/k} \qquad (1 \le i \le 4).$$

Here M_0 is the odd-contact radical for $\Theta = \{0, -1/2, -1, 1/2399, -32/407\}$.

Proof. The product is positive in both embeddings, so $C_0 > 0$. The fractional linear map $t \mapsto 2 + 1/t$ identifies the five contacts with the original five, outside the exceptional primes. Consequently $h(t) \le (H + \log 3)M_0(t)^\eta$. Put $T = (H + \log 3)|N|^{\eta/k}$. The unscaled detector and each scaled detector give $h(A_j) \le 2T$. Hence $h(C_0) \le T$. For $C_1 = u/v$ in lowest terms with $v > 0$, $h(1 + C_1^2) = \log(u^2 + v^2) \ge 2h(C_1)$, so $h(C_1) \le T/2$. The elementary height inequality in the fixed basis gives

$$h(C) \le \tfrac{3}{2}T + \tfrac{3}{2}\log 2 < 7H|N|^{\eta/k}.$$

Lemma 4.2 bounds every summand by $h(C)$. □

4.4 Point charts with no free parity witnesses

Lemma 4.4. *Suppose $R = (a, b) \in E(K)$ is not a two-torsion point. At least one of $R, R + T_0$ has nonzero rational component of its ordinate. Thus every nonzero double is represented by $2R$ in the chart below, and every odd multiple $[n]P$ with $n \neq 1$ is represented by $P + 4R$ in the same chart.*

Proof. If both ordinates had zero rational component, then $b \in \sqrt{2}\mathbb{Q}^\times$ and $D/a^2 = b(R + T_0)/b(R) \in \mathbb{Q}$. Therefore $a/d \in \mathbb{Q} \cup \sqrt{2}\mathbb{Q}$. Since $b^2 = d^3(a/d)((a/d)^2 - 1) \in \mathbb{Q}^\times$, this would force $d^3 \in \mathbb{Q} \cup \sqrt{2}\mathbb{Q}$, contrary to $d^3 = 45 - 29\sqrt{2}$. Translation by T_0 preserves both $2R$ and $4R$.

If n is odd and $n \neq 1$, take $R = [(n - 1)/2]R_0$, so $P + 4R = [n]P$. Its $4R$ is nonzero. The affine addition denominator $x(4R) - x(P)$ cannot vanish: otherwise $4R = \pm P$, whereas $4E(K) = 2\mathbb{Z}P$ contains only even multiples. The same translation by T_0 ensures the chosen chart is defined. □

For each of the nine names $A, S, T, Z, H_0, U_A, U_S, U_T, U_Z$, retain a rational pair (a_0, a_1). Write $a = a_0 + a_1\sqrt{2}$ and $a^3 - Da = A' + B'\sqrt{2}$. Introduce a finite algebra root Y with

$$Y^4 - 4A'Y^2 + 8(B')^2 = 0, \qquad b = Y/2 + (B'/Y)\sqrt{2}.$$

Require $Y \neq 0$, $(A')^2 - 2(B')^2 \neq 0$, and $4Y^3 - 8A'Y \neq 0$. Then $R = (a, b) \in E(K)$. For the four root names set the output point to $P + 4R$; for the other five set it to $2R$. All arithmetic is by the usual affine addition and duplication formulas with their denominators guarded. The derivative condition is automatic at rational points of this chart but is retained to ensure geometric simplicity in the finite algebra. Indeed, since $Y \neq 0$, a vanishing derivative would give $Y^2 = 2A'$; the quartic would then give $(A')^2 = 2(B')^2$, forcing $A' = B' = 0$ over $\mathbb{Q}$, contrary to the norm guard. It introduces no extra coordinate.

In particular, put $C_S = S - P = 4R_S$. This point, rather than S, will be used in the scalar and size certificates. Write

$$(A, S, T, Z) = ([n]P, [a]P, [t]P, [b]P).$$

The four labels are nonzero odd integers; the labels are consequences of the group hypothesis, not additional quantified variables.

4.5 The rational system

Retain the input X, the eighteen point-chart coordinates, one rational λ, two parameters for each of three comparison poles, and two parameters for a scalar pole. There are

$$1 + 18 + 1 + 6 + 2 = 28$$

retained rational coordinates. Set

$$\beta = 216X^4 + 450459.$$

The following elementary input map eliminates every scalar correction choice.

Lemma 4.5. *For rational X, the number $\beta = 216X^4 + 450459$ is integral if and only if X is integral. It is greater than 2401. For integral X, its contact $r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1)$ is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, with $v_2(r) = v_3(r) = 2$ and $v_{17}(r) \in \{0, 2\}$.*

Proof. This is proved, with its complete residue table and exceptional valuation calculation, in Lemmas 9.1 and 9.2. □

Put $Q_0 = P$, $(Q_1, Q_2, Q_3, Q_4) = (A, S, T, Z)$, $U_0 = H_0$, and $V_i = U_i - Q_i$ for $0 \le i \le 4$. For $k \in \{\rho_0, \rho_1\}$ define

$$\ell_k(Q) = z(Q) \sum_{j=0}^{k/4-1} c_j (Df(Q)^2)^j, \quad c_0 = 1, \quad c_1 = -2/5, \quad c_j = c_{j-1} \frac{4j-5}{4j+1}.$$

Put $(k_1, k_2, k_3, k_4) = (\mu, \kappa, \kappa, \kappa)$ and $(r_1, r_2, r_3, r_4) = (\rho_0, \rho_1, \rho_1, \rho_1)$. Set

$$L_i = \frac{\ell_{r_i}(U_i)}{\ell_{\rho_1}(H_0)}, \qquad L_i' = \sigma\left(\frac{\ell_{r_i}(V_i)}{\ell_{\rho_1}(V_0)}\right).$$

The sixteen field-valued tests, with their orders, are

$$\begin{array}{c|c} f(U_0), \ldots, f(U_4), \sigma f(V_0), \ldots, \sigma f(V_4) & 1 \text{ each} \\ L_i - L_i' & k_i \\ L_1 - \beta & \mu \\ L_3^2 - L_2(L_2-2)L_3L_4 + (L_2-2)^2 L_4^2 - 1 & \kappa. \end{array}$$

Denote these tests by D_j and their orders by m_j. Use precisely the three scales of Corollary 4.3. For each such A_r put

$$t_r = A_r \lambda, \quad \delta_r = t_r - 1, \quad F_r = \prod_{\theta \in \Theta} (t_r - \theta), \quad R_r = \frac{F_r}{A_r^5 \delta_r^6},$$

$$W_r = \prod_{j=1}^{16} \left(1 + R_r \left(\frac{D_j}{\delta_r^{m_j+1} R_r^{m_j}}\right)^2\right), \qquad b_r = \frac{R_r}{\mathrm{Tr}_{K/\mathbb{Q}}(W_r^{-1})^2}.$$

All F_r, δ_r, norms of W_r, and displayed denominators are required to be nonzero. In the coefficient DAG, if $W_r = w_0 + w_1\sqrt{2}$, the last expression is exactly

$$b_r = \frac{F_r (w_0^2 - 2w_1^2)^2}{4 A_r^5 \delta_r^6 w_0^2}.$$

For each nonzero pole input b use two retained parameters τ, s and a fixed finite choice $\varepsilon \in \{0, 1\}$. Set

$$B = 17b, \quad e = \frac{\tau^2 + (4\varepsilon - 1)B}{\tau^2 + 3B}, \quad h_0 = e(B - b^2), \quad q^2 = \frac{(s^2 - Bh_0)(s^2 + 3Bh_0)}{B}.$$

Require $\tau^2 + 3B \neq 0$. The first original conic is recovered by $p = -4B\tau(1-\varepsilon)/(\tau^2 + 3B)$. Conversely every point of that conic is covered: if $e \neq 1$ take $\tau = p/(e-1)$ and $\varepsilon = 0$; if $e = 1$ take $\varepsilon = 1, \tau = 0$. This includes $e = -1/3, p = 0$. There is no nonzero condition on τ, p, or q.

For the fourth, scalar pole put

$$r_\beta = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1), \quad \delta_\beta = 1 + (17 r_\beta)^2, \quad R_\beta = \frac{102^2 r_\beta}{\delta_\beta^2}.$$

Require $R_\beta \neq 0$. Write the rational components of $f(H_0), f(C_S), z(C_S)/z(H_0)$ as $g_0, \ldots, g_5$, in that order. For $N_R(u, v) = u^2 + Rv^2$, set

$$w_0 = N_R(1, g_0/R), \quad w_1 = N_R(g_1/R, g_2/R), \quad w_2 = g_3/R, \quad w_3 = N_R(g_4/R^m, g_5/R^m),$$

$$W = N_R(N_R(w_0, w_1), N_R(w_2, w_3)), \qquad R = R_\beta.$$

Require $W \neq 0$, and give $R_\beta W^2$ the same pole certificate. There is no scalar correction parameter. The only original finite choices are the four conic bits, hence 2^4 source branches. For

rational input $R_\beta > 0$, the constant-left norm tree in fact has $W \geq 1$, so this nonzero condition is automatic; it is harmless and convenient to retain the same denominator-guard convention.

For the finite-anchor profile require

$$\prod_{j=1}^{M} \mathrm{Norm}_{K/\mathbb{Q}}(x(C_S) - x([j]P)) \neq 0, \qquad M = 2^{2093449}.$$

Clear denominators factor by factor using the canonical primitive triple $x([j]P) = (a_j + b_j\sqrt{2})/c_j$, $c_j > 0$. If $x(C_S) = (N_0 + N_1\sqrt{2})/D_0$, the integer factor is

$$(c_j N_0 - a_j D_0)^2 - 2(c_j N_1 - b_j D_0)^2.$$

The multiples of the fixed point are computed by exact binary elliptic multiplication. This is a finite, fully indexed polynomial guard, although its coefficients are not expanded.

4.6 Local implications and soundness

We include the local argument to make the effect of direct targets explicit. At a prime $p > 76176$ where $v_p(R)$ is odd, consider $W = \prod_j (1 + R(D_j/R^{m_j})^2)$, $T = \mathrm{Tr}(W^{-1})$, and a pole certificate for R/T^2. Each factor has valuation

$$\min\{0, v_p(R) + 2(v_\mathfrak{p}(D_j) - m_j v_p(R))\},$$

since its two terms have valuations of different parity. If $\lambda = \max_{\mathfrak{p}|p} v_\mathfrak{p}(W) \leq 0$, then $v_p(T) \geq -\lambda$. The pole implication gives

$$0 < v_p(R/T^2) \leq v_p(R) + 2\lambda.$$

Thus $v_p(R) > 0$ and $\lambda > -v_p(R)/2$. At a place attaining the maximum, every factor valuation is at least λ, whence $v_\mathfrak{p}(D_j) > (m_j - 3/4)v_p(R) \geq m_j - 3/4$ and therefore $v_\mathfrak{p}(D_j) \geq m_j$. For the normalized detectors, δ_r is a unit at each consumed contact prime, so this argument applies to the original D_j.

Write $H_0 = [h]P$ and $U_i = [u_i]P$. The first ten tests place the two families in the corresponding formal groups. Formal logarithms have exact ratios u_i/h and $(u_i - n_i)/(h-1)$. The displayed truncations have relative precision k_i: the omitted series terms start with valuation $k_i v(z)$, and the factorial bound on their denominators gives the same conclusion for all subsequent terms at $p > 76176$. At least one of $h, h-1$ is a p-unit. The associated ratio is integral; the cross-test of order k_i makes both ratios integral. Relative precision is therefore absolute precision. Multiplication by h and $h-1$ and subtraction gives

$$L_i \equiv n_i \pmod{p^{k_i}}.$$

This argument also holds at an inert prime. Integer polynomial congruences then give the corresponding congruence for the direct Lucas residual.

The scalar certificate gives

$$M_5(\beta)^m \mid (a-1), \qquad h(\beta) \leq H|a-1|^{\eta/m}.$$

Indeed negative valuations of $\beta = 216X^4 + 450459$ are multiples of four outside the fixed primes two and three. At every consumed finite odd contact, R_β has positive odd valuation ν; its normalizer is a unit and its fixed multiplier is a square unit. For N_R there is no cancellation between the two terms. If a leaf at depth j contributes to the root, its valuation has the form $2^j v(\text{leaf}) + e\nu$, $0 \leq e < 2^j$. The pole implication forces the root valuation to exceed $-\nu/2$; hence every normalized leaf has valuation greater than $-\nu$. An original test assigned order m consequently has valuation at least $(m-1)\nu + 1 \geq m$. The order-one tests put H_0, C_S in

the formal group; the order-m ratio implies $v_p((a-1)/h) \geq m$, and thus $p^m \mid a-1$. The contact-height hypothesis proves the stated height bound.

Set $c = |a-1|$, $B = \max(|n|, |a|, |t|, |b|)$ and $\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1$. If $\Delta \neq 0$, the local implications and the three product-height scales give

$$B^2 - 6 < 7H(5B^4)^{\eta/\kappa}.$$

The index size supplied by either profile contradicts this inequality, as proved in Section 6. Hence $\Delta = 0$ and $\log B > |a| - 3 \geq c - 4$. If $\beta \neq n$, let N be the absolute value of the numerator of $\beta - n$ in lowest terms. It satisfies $N \leq 2e^{h(\beta)}B$, and each contact radical has its μth power dividing N. Therefore

$$B^2 - 6 < 7H(2e^{h(\beta)}B)^{\eta/\mu}.$$

The scalar estimate and the same profile inequalities again contradict Lucas growth. Thus $\beta = n \in \mathbb{Z}$, and Lemma 4.5 implies $X \in \mathbb{Z}$.

4.7 Simultaneous completeness

Fix $X \in \mathbb{Z}$. Lemma 4.5 gives an odd integer $\beta > 2401$, with r_β positive and square at the three preparation primes. The integer $\delta_\beta = 1 + (17r_\beta)^2$ is a unit at two and three because r_β is divisible by both, and at seventeen because of its factor seventeen. Consequently R_β has square reduced denominator, is square locally, and has valuations at least $4, 4, 2$ at $2, 3, 17$. Moreover

$$0 < R_\beta = \frac{102^2 r_\beta}{(1 + (17r_\beta)^2)^2} < \frac{102^2}{17^4} < 1.$$

No finite squareclass adjustment is required.

Choose a real compact neighborhood for H_0 near, but away from, the identity in both embeddings, with $f(H_0)$ small and $z(H_0)$ bounded away from zero. The positive orbit of every suitable finite-index subgroup of $\mathbb{Z}P$ is dense in the identity component of its real closure. Thus one may choose a positive h_* divisible by 64, and deep in every formal group over the finite support of R_β, whose point lies in this neighborhood. Choose a positive integer s divisible by six, still deeper at these places, sufficiently large for the selected size certificate, and with $[s]P$ sufficiently close to the real identity. Set $a = s + 1$, so a is odd and $a \equiv 1 \pmod 3$. Take $S = [a]P$, and use $C_S = [s]P$ in the scalar certificate. The normalized leaves of its norm tree are then small in the real absolute value and sufficiently deep locally, uniformly for H_0 in a smaller neighborhood. Its root is a local unit and $|W - 1| < 1/2$.

Odd Lucas labels are available explicitly. Let $U_0 = 0$, $U_1 = 1$, $U_{j+1} = aU_j - U_{j-1}$, and put

$$k = 2(a-2), \qquad b = U_k/(a-2), \qquad t = U_{k+1}.$$

Modulo $a-2$ one has $U_j \equiv j$, so b is integral. Modulo two the sequence has period $0, 1, 1$. Since $a \equiv 1 \pmod 3$, $k \equiv 1 \pmod 3$, so b, t are odd. They are positive and greater than one. The identity $U_{k+1}^2 - aU_kU_{k+1} + U_k^2 = 1$ gives $\Delta = 0$. The four root labels β, a, t, b are therefore all odd and at least three, and Lemma 4.4 covers their points.

Temporarily set $H_0 = [h]P$ and $U_i = [hn_i]P$. Local continuity provides a progression $h \equiv h_* \pmod L$, with $64 \mid L$, preserving the scalar units and fixed lower bounds for all comparison tests at $2, 3, 17$. Include the order of the finite component group of the real closure in L. Remove the finitely many comparison pole loci from the real neighborhood and pass to a compact subset; the tests are then uniformly bounded there. These choices precede the prime-pattern application.

Choose D_0 with $D_0^2\theta/A_r \in \mathbb{Z}$ for every contact and scale, put $v = D_0^2w^2$, and choose $\lambda = u/v$. The contact numerators $u - (\theta/A_r)v$ are a finite set of distinct prime forms after repetitions are removed. Apply the simultaneous prime-pattern hypothesis beyond all fixed bad primes, with enough divisibility of w and a sufficiently large lower bound on u/w^2. The curve has full

rational two-torsion at every selected residue field. Because P is a double, $d_{\pm}(p)/4$ annihilates its reduction. Define

$$U = \operatorname{lcm}_p(d_-(p)/4), \qquad V = \operatorname{lcm}_p(d_+(p)/4).$$

The order and avoidance hypotheses give $\gcd(U, V) = 1$, $\gcd(U, L) = 16$, and $\gcd(V, L) = 1$. Thus

$$h \equiv 0 \pmod{U}, \qquad h \equiv 1 \pmod{V}, \qquad h \equiv h_* \pmod{L}$$

are compatible. Their progression still has the required real closure, so H_0 can lie in the chosen compact set. At one place over each contact prime the points H_0, U_i are formal; at the conjugate place V_0, V_i are formal. Both exact logarithmic ratios are n_i. All sixteen tests therefore have their assigned orders, including the direct Lucas polynomial since $\Delta = 0$.

For completeness of each comparison pole, note the exact identity

$$R_r = \frac{\prod_{\theta \in \Theta}(u - (\theta/A_r)v)}{v^5(A_r\lambda - 1)^6}.$$

The denominator is a rational square. At the five contact primes $v_p(R_r) = 1$. At either place over such a prime each inverse norm factor has residue zero or one, and at the successful place every residue is one. Its reciprocal trace is thus a unit with residue one or two. Consequently b_r has positive odd valuation only at these primes and has square reduced denominator. Their prescribed congruences make b_r square at $2, 3, 17$.

Uniformly on the compact real set, $R_r \sim A_r^{-6}\lambda^{-1}$, $W_r = 1 + O(\lambda^{-3})$, and $b_r \sim (4A_r^6\lambda)^{-1}$. Hence $0 < b_r < 17$ for a sufficiently large prescribed lower bound on λ. At $p \in \{2, 3, 17\}$ put $\nu = v_p(v)$ and $\eta = v_p(A_r)$. Sufficient divisibility of w gives $v_p(R_r) = \nu - 6\eta$. The nonconstant terms of the norm factors have valuations at least $3\nu + (10m_j - 8)\eta - 2c_p$, where c_p is the previously fixed lower-bound constant. Their inverse trace therefore has valuation $v_p(2)$, and $v_p(b_r) = \nu - 6\eta - 2v_p(2)$. These valuations can exceed every required preparation threshold. The pole hypothesis supplies the three comparison certificates.

The scalar pole input retains square denominator and the specified local squareclasses because W is a unit at every prime in its fixed support. Also $0 < R_\beta W^2 < 4 < 17$. The same pole hypothesis supplies its certificate with preparation zero. All guards are met, and the conic parametrization covers all these witnesses. This proves completeness for every integer input.

4.8 The finite polynomial and its bounds

The thirteen formal roots consist of nine point quartics and four pole quadratics. A rational equation $q^2 = N/D$ with $D \neq 0$ is made monic by $Z = Dq$, so $Z^2 - ND = 0$. Its zero-radicand branch is $Z = 0$ with $ND = 0$; its other branch has $Z^2 = ND$ and $Z \neq 0$. Together with the point derivative guards this makes every admitted geometric point simple. The rank is at most

$$4^9\, 2^4 = 2^{22}.$$

Theorem 3.1 eliminates all thirteen roots using one new rational variable U. For completeness, its literal recipe is as follows. On each quadratic support branch, put $m = 13$ and let $n = 4^9 2^k$, where k is the number of nonzero quadratic branches. Let G be the product of the source guards and E_i the closed zero-radicand conditions. For every $0 \leq \ell \leq mn(n+1)/2$ and $1 \leq d \leq n$, form

$$\lambda_\ell = 1 + \sum_{i=1}^{m} \ell^i z_i, \quad H(V) = \det(I - VM_{G\lambda_\ell}) = \sum a_j V^j, \quad R_i(V) = \operatorname{Tr}(M_{E_i} \operatorname{adj}(I - VM_{G\lambda_\ell})).$$

Put $H_d = \sum_{j=0}^{d} a_j V^j$, $g = a_d \operatorname{Res}(H_d, H_d')$, $L = a_d^2 g$, and take the factor

$$\left(1 + \sum_{j=1}^{d} a_{d-j} a_d^{2j-1} g^j U^j\right)^2 + \sum_{j>d} a_j^2 + \sum_i \left((LU)^{n-1} R_i(1/(LU))\right)^2.$$

Coefficient reversal makes the last expressions polynomials at $LU = 0$. Multiply these factors over the displayed ranges, the 2^4 quadratic masks, and all 2^4 original finite choices. This is a completely specified integer polynomial in the retained coordinates and U, with no formal root left as a free variable.

For the semilocal profile retain two additional rational coordinates v, w. With $f(C_S) = f_0 + f_1\sqrt{2}$, put $F_0 = f_0^2 + f_0 f_1 - f_1^2$ and $C = 3^{2^{26}}$, and introduce the finite quadratic root

$$u^2 = 3F_0^2 - 6v^2 + 2w^2 - 2C^4.$$

Lemma 7.2, together with the binary-form valuation in Lemma 7.1, identifies this with $f_0/C, f_1/C \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. Proposition 8.1 supplies the size bound used in the profile proof. Its zero root is included by the same quadratic support split. For this profile the finite recipe has $m = 14$, five quadratic support bits, and branch rank $n = 4^9 2^k$ with $0 \le k \le 5$; the four original conic bits are unchanged.

Theorem 4.6. *Under the stated arithmetic hypotheses, the two specified polynomials P_{29} and P_{31} have rational input projection exactly $\mathbb{Z}$. Their declared total numbers of variables, including the input, and certified bounds are*

	finite algebra rank	*degree bound*	*coefficient-height bound*
P_{29}	2^{22}	$2^{2093605}$	$2^{2^{6280501}}$
P_{31}	2^{23}	2^{180}	$2^{2^{184}}$.

The coefficient height is the largest absolute value of an integer coefficient. The displayed degrees and heights are upper bounds, not claims of exact expanded values.

Proof. The local implications and profile size calculations prove soundness. The simultaneous construction proves completeness. Every remaining formal root has a monic equation, and all rational denominators are explicitly guarded. The finite algebra theorem therefore preserves the rational input projection and uses one final coordinate beyond the retained 28 or 30. The coefficient DAGs have respectively 6386 and 4367 nodes. Section 6.4 proves the degree and coefficient-height bounds from the finite algebra and the fixed-point exclusion. □

The finite-anchor polynomial is specified by an exact indexed product with canonically computed fixed coefficients. The semilocal polynomial has no such product. Both use finite determinant and coefficient operations; the construction gives complete polynomial specifications without requiring expanded monomial lists.

5 Quadratic height bounds for the fixed elliptic multiples

Put $K = \mathbb{Q}(\sqrt{2})$ and

$$D = 11 - 6\sqrt{2}, \qquad E:\ y^2 = x^3 - Dx, \qquad P = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32}\right).$$

The height h below is the absolute logarithmic Weil height. If $x = (a + b\sqrt{2})/c$ with $a, b, c \in \mathbb{Z}$, $c > 0$, and $\gcd(a, b, c) = 1$, put

$$h_0(x) = \lceil \log_2 \max(1, |a|, |b|, c) \rceil .$$

Lemma 5.1. *The point P belongs to $E(K)$ and has infinite order.*

Proof. Substitution verifies the curve equation. At the unique place above two, normalize v by $v(2) = 1$. The displayed coordinates have valuations $v(x(P)) = -3$ and $v(y(P)) = -9/2$, so the formal parameter $z = -x/y$ has $v(z(P)) = 3/2$. The integral formal group of this Weierstrass equation is odd: for every integer m, $[m](Z) = mZ + Z^3 g_m(Z^2)$ with integral coefficients. At a parameter of valuation at least $3/2$, multiplication by an odd integer preserves its valuation, and multiplication by two raises it by one. In each case the leading term has strictly smaller valuation than the remaining terms. Thus

$$v(z([n]P)) = 3/2 + v_2(n) \qquad (n \neq 0).$$

It is finite, and hence $[n]P \neq \mathcal{O}$ for every nonzero n. In particular all its abscissas are finite and nonzero. □

Lemma 5.2. *For $n \geq 1$, write $h_n = h(x([n]P))$. Then*

$$h_{2n} \leq 4h_n + \log 392,$$
$$h_{2n+1} \leq 2h_n + 2h_{n+1} + \log 8036.$$

The second formula is used for $n \geq 1$.

Proof. The duplication and differential-addition identities are

$$x(2Q) = \frac{(x(Q)^2 + D)^2}{4x(Q)(x(Q)^2 - D)}, \qquad x(Q+R)x(Q-R) = \left(\frac{x(Q)x(R) + D}{x(Q) - x(R)}\right)^2.$$

For the second identity, the usual addition formulas reduce it to

$$(u+v)^2(uv - D)^2 - 4uv(u^2 - D)(v^2 - D) = (u - v)^2(uv + D)^2,$$

which is a polynomial identity. Taking $Q = [n]P$ and $R = [n+1]P$ makes $x(Q - R) = x(P)$. Its denominator is nonzero: an equality $x([n]P) = x([n+1]P)$ would contradict the infinite order of P.

For each place, bound a homogeneous polynomial by the largest coordinate norm to its degree. At a finite place the coefficient bound is the maximum coefficient norm; at an infinite place use the sum of the coefficient norms. Applied to duplication, whose homogeneous numerator and denominator have degree four, this gives

$$h(x(2Q)) \leq 4h(x(Q)) + 2h(D) + \log 8.$$

Applied to differential addition, with bidegree $(2, 2)$ in the two abscissas and denominator $x(P)(x(Q) - x(R))^2$, it gives

$$h(x(Q+R)) \leq 2h(x(Q)) + 2h(x(R)) + 2h(D) + h(x(P)) + \log 4.$$

For example, the archimedean coefficient sums for the second formula are bounded by $4\max(1, |D|)^2 \max(1, |x(P)|)$, and the corresponding finite-place bound has no factor four. These local estimates sum to the stated absolute-height estimates.

Both real conjugates of D exceed one, and $N_{K/\mathbb{Q}}(D) = 49$, so $h(D) = \log 7$. Both real conjugates of $x(P)$ exceed one, its norm is $41^2/8^2$, and its denominator ideal has norm 8^2. Consequently $h(x(P)) = \log 41$. Substitution gives $2h(D) + \log 8 = \log 392$ and $2h(D) + h(x(P)) + \log 4 = \log 8036$. □

Theorem 5.3. *For every integer $n \geq 1$,*

$$h(x([n]P)) \leq n^2 \log 23534 - \log 574, \qquad h_0(x([n]P)) \leq 30n^2.$$

These inequalities are unconditional for the displayed curve and point.

Proof. Write $A = \log 23534$ and $B = \log 574$. The integer identities

$$23534 = 41 \cdot 574, \qquad 574^3 = 23534 \cdot 8036, \qquad 574^3 \geq 392$$

give $h_1 = A - B$, $3B = A + \log 8036$, and $3B \geq \log 392$. Induction using Lemma 5.2 proves $h_n \leq An^2 - B$. For an even index,

$$h_{2n} \leq 4(An^2 - B) + \log 392 \leq A(2n)^2 - B.$$

For an odd index at least three,

$$\begin{aligned} h_{2n+1} &\leq 2(An^2 - B) + 2(A(n+1)^2 - B) + \log 8036 \\ &= A(2n+1)^2 + A - 4B + \log 8036 = A(2n+1)^2 - B. \end{aligned}$$

We next compare coefficient height with absolute height. Write $x = (a + b\sqrt{2})/c$ in the primitive form above, and let $\mathfrak{d} = \{u \in \mathcal{O}_K : ux \in \mathcal{O}_K\}$ be its denominator ideal. Since $\mathcal{O}_K = \mathbb{Z}[\sqrt{2}]$, the integer c is the least positive rational integer with $cx \in \mathcal{O}_K$. At a prime above a rational prime p, put $d_{\mathfrak{p}} = \max(0, -v_{\mathfrak{p}}(x))$. The p-adic exponent of $N\mathfrak{d}$ is $\sum_{\mathfrak{p}|p} f_{\mathfrak{p}} d_{\mathfrak{p}}$. Multiplying by a ramification index $e_{\mathfrak{p}}$ makes this at least $d_{\mathfrak{p}}$. Therefore $(N\mathfrak{d})x$ is integral and $c \mid N\mathfrak{d}$. If $h_{\mathrm{fin}}(x)$ denotes the finite-place part of $h(x)$, then $N\mathfrak{d} = \exp(2h_{\mathrm{fin}}(x))$.

Writing x' for the real conjugate, one has

$$|a| = c\left|\frac{x + x'}{2}\right| \leq c \max(|x|, |x'|), \qquad |b| = c\left|\frac{x - x'}{2\sqrt{2}}\right| \leq c \max(|x|, |x'|).$$

The product $\max(1, |x|)\max(1, |x'|)$ equals $\exp(2h_\infty(x))$. It follows that

$$\max(1, |a|, |b|, c) \leq \exp(2h(x)).$$

Finally $23534 < 2^{15}$, so the first height bound gives $\exp(2h_n) < 2^{30n^2}$ and hence $h_0(x([n]P)) \leq 30n^2$. □

5.1 The indexed exclusion polynomial

Let $\mathcal{A}$ be a finite free algebra over an integer polynomial ring, with a fixed basis. For a multiplication matrix use the maximum column sum of the coefficient ℓ^1-norms. This norm is submultiplicative. Suppose $b \geq 0$ and the multiplication matrices of $N_0, N_1, D_0 \in \mathcal{A}$ have norm at most 2^b. Write $x([n]P) = (a_n + b_n\sqrt{2})/c_n$ in primitive form and set

$$\mathcal{E}_n = (c_n N_0 - a_n D_0)^2 - 2(c_n N_1 - b_n D_0)^2, \qquad \mathcal{E}_{\leq M} = \prod_{n=1}^{M} \mathcal{E}_n.$$

Each fixed triple is computed by exact elliptic arithmetic. The product is finite and is a literal polynomial specification, independently of whether its coefficients are expanded.

Proposition 5.4. *For every integer $M \geq 1$,*

$$\|M_{\mathcal{E}_{\leq M}}\| \leq 2^{20M^3 + 30M^2 + (2b+14)M} \leq 2^{(2b+64)M^3}.$$

If $b \leq 1048$ and $M \geq 64$, the first exponent is at most $21M^3$.

Proof. Theorem 5.3 and the triangle inequality give

$$\|M_{c_n N_0 - a_n D_0}\|,\ \|M_{c_n N_1 - b_n D_0}\| \le 2^{30n^2+b+1}.$$

Squaring and summing the two terms of $\mathcal{E}_n$ therefore gives $\|M_{\mathcal{E}_n}\| \le 2^{60n^2+2b+4}$. Sum these logarithmic bounds and use $\sum_{n=1}^{M} n^2 = M(M+1)(2M+1)/6$. The second bound follows from $M, M^2 \le M^3$. For the last assertion, $2b + 14 \le 2110$, and

$$30 \cdot 64 + 2110 = 4030 < 64^2.$$

The function $M^2 - 30M$ is increasing for integers $M \ge 64$. Thus $30M^2 + (2b+14)M \le M^3$, proving the result. □

At a rational specialization with $D_0 \neq 0$, the factor $\mathcal{E}_n$ vanishes exactly when $(N_0 + N_1\sqrt{2})/D_0 = x([n]P)$. Indeed a rational pair satisfies $u^2 - 2v^2 = 0$ only when both entries vanish. Thus changing this coefficient bound changes neither the exclusion polynomial nor its rational zeros.

For the 29-variable specification, the recorded coordinate bound is $b = 1048$, and $M = 2^{2093449}$. The indexed multiplication matrix therefore has logarithmic norm at most

$$21M^3 < 2^{6280352}.$$

The independently checked determinant, reciprocal, trace and product bounds consequently give

$$\deg P_{29} \le 2^{2093605}, \qquad H(P_{29}) \le 2^{\,2^{6280501}}.$$

The source polynomial is specified by its finite-algebra relations, branches, guards, and the indexed exclusion product above. Its 29 variables include the distinguished input. The 31-variable specification does not contain this exclusion product and retains its bounds $\deg P_{31} \le 2^{180}$ and $H(P_{31}) \le 2^{\,2^{184}}$.

6 Trading divisibility order for a shorter size exclusion

The number of factors in the finite size exclusion has a much larger effect on the final degree than the powers in the scalar norm tree. The two profiles use the explicit arithmetic input [1, Theorem `thm:quartic-height19`]:

$$h(t) \le H M_5(t)^{19}, \qquad 1 \le H < 2^h, \qquad h = 2093448.$$

Here H is the explicit integer $2^{110}210^{1800}(\prod_{p\le 76176} p)^{19} + 128$, not an unspecified constant. Trial division enumerates exactly 7499 primes in that product, the largest being 76163. Their product has binary length 109445, and exact integer substitution gives binary length 2093448 for H. The supplied integer checker recomputes these values without a supplied prime list or sieve. The enumeration is complete because each candidate is divided by every previously found prime not exceeding its square root; the usual least-prime-divisor argument proves this criterion by induction. The definition of binary length then gives the displayed strict bound. It implies the same inequality with exponent 24. We use that weaker exponent for the finite-anchor construction below, and exponent 19 directly for the smaller-degree semilocal alternative. No assertion about additional prime separation is needed.

6.1 A general parameter calculation

Let the contact-height exponent be η. Assign order κ to the three Lucas cross-tests and to the Lucas polynomial, order μ to the input cross-test and input comparison, and order m to the two scalar ratio components. The ten formal-membership tests retain order one. Choose formal logarithm truncations of relative precision at least the assigned comparison orders. The local, conic and norm-tree arguments of Section 4 are unchanged. If

$$c = |a-1|, \qquad B = \max(|n|, |a|, |t|, |b|),$$

they give

$$h(\beta) \le Hc^{\eta/m}, \tag{1}$$

$$\Delta \neq 0 \quad \Longrightarrow \quad B^2 - 6 < 7H(5B^4)^{\eta/\kappa}, \tag{2}$$

$$\beta \neq n \quad \Longrightarrow \quad B^2 - 6 < 7H(2e^{h(\beta)}B)^{\eta/\mu}. \tag{3}$$

For the second implication, Δ is the integral Lucas residual; for the third, use the numerator of $\beta - n$ in lowest terms. The third implication is needed only after the second has forced $\Delta = 0$, at which point

$$\log B > |a| - 3 \ge c - 4.$$

An increase of m does not require a longer formal logarithm series. The scalar test is the valuation of $z(C_S)/z(H_0)$ itself. For formal points at the consumed primes, the logarithm divided by the local parameter is a unit. Thus

$$v_{\mathfrak{p}}(z(C_S)/z(H_0)) = v_p((a-1)/h_0),$$

where $H_0 = [h_0]P$. This yields $p^m \mid a-1$ directly. Larger m only asks for deeper choices at finitely many places in completeness. Since m is fixed, the local continuity and Chinese remainder argument still applies. Powers such as R_β^m are exact integer-power operations in the coefficient DAG.

Lemma 6.1 (A cutoff linear in the height constant)**.** *Suppose $h \ge 20$ is an integer and $1 \le H < 2^h$. Work with the contact-height exponent $\eta = 24$, and set*

$$E = h + 1, \quad M = 2^E, \quad m = 48E, \quad \kappa = 128, \quad \mu = 24.$$

If $c > M$, the three implications (1)–(3) force $\Delta = 0$ and $\beta = n$.

Proof. Since c and a are integers, $|a| \ge c - 1 \ge M$. Because $24/m = 1/(2E)$ and the exponent $1/(2E) - 1$ is negative,

$$\frac{Hc^{24/m}}{c} < 2^h M^{1/(2E)-1} = 2^{-1/2} < \frac{3}{4}.$$

Consequently $h(\beta) < 3c/4$.

If $\Delta \neq 0$, then $B \ge M > \sqrt{12}$ and

$$\tfrac{1}{2}B^2 < B^2 - 6 < 7\,5^{3/16} H B^{3/4} < 14 H B^{3/4}.$$

Thus $B^{5/4} < 28H$. On the other hand

$$B^5 \ge 2^{5h+5} > 28^4\, 2^{4h} > (28H)^4,$$

since $h \ge 20$ and $28^4 < 2^{20}$. This is impossible, and $\Delta = 0$.

If $\beta \neq n$, equation (3) becomes $B^2 - 6 < 14He^{h(\beta)}B$. Since $B, H, e^{h(\beta)} \ge 1$,

$$B < 20He^{h(\beta)}, \qquad \log B < \log(32H) + h(\beta).$$

For $h \geq 5$ one has $(h+5)\log 2 < 2^{h-2}$, as follows at $h = 5$ by direct comparison and thereafter by induction. Therefore

$$\log(32H) < 2^{h-2} = M/8 < c/8, \qquad \log B < 7c/8 < c - 4,$$

the last inequality using $c > M \geq 2^{21}$. This contradicts Lucas growth. Hence $\beta = n$. □

For $h = 2093448$ the parameters are

$$M = 2^{2093449}, \qquad m = 100485552, \qquad \kappa = 128, \quad \mu = 24.$$

The longest formal logarithm truncation is 128, so the inherited $p > 76176$ estimate applies with ample room. The first comparison uses truncation 24. Both directions of the rational source proof are therefore valid with these parameters. In particular the large scalar exponent introduces no extra free coordinate, finite generator, finite branch, or arithmetic hypothesis.

6.2 Small test orders for the semilocal variant

For a fixed 31-variable semilocal presentation, the literal anchor constant has degree zero. One can therefore reduce the degree by decreasing the test powers and increasing that fixed constant. Use the contact-height exponent 19 directly and assign

$$\kappa = 39, \qquad \mu = 10, \qquad m = 20.$$

Use formal truncation 40 for the Lucas comparisons and denominators, and truncation 12 for the input comparison. A truncation precision need only exceed the assigned divisibility order; they need not be equal. The inequality $\kappa > 2 \cdot 19$ is what makes (2) subquadratic in B, and $\mu > 19/2$ is what makes (3) useful. These are the smallest integer orders satisfying those two strict inequalities. This is a statement about this parameter calculation, not an optimality theorem for all Diophantine definitions.

Choose the semilocal constant $C = 3^{2^{26}}$. Proposition 8.1 gives

$$c \geq 2\,3^{2^{25}-1} > (64H)^{20}.$$

For the displayed h, this follows from $3 > 2^{3/2}$ and the exact integer comparison

$$1 + \frac{3}{2}(2^{25} - 1) > 20(h + 6).$$

Hence $h(\beta) \leq Hc^{19/20} < c/64$.

If $\Delta \neq 0$, the inequality $5^{19/39} < \sqrt{5} < 16/7$ gives

$$B^2 - 6 < 16HB^{76/39}.$$

As $B \geq |a| \geq c - 1 > \sqrt{12}$, it follows that $B^{2/39} < 32H$, and thus $B < (32H)^{39/2}$. But $|a| \geq c - 1 > c/2 > (64H)^{20}/2 > (32H)^{39/2}$, a contradiction. We again obtain $\Delta = 0$.

If $\beta \neq n$, then

$$B^2 - 6 < 7\,2^{19/10} H e^{19h(\beta)/10} B^{19/10}.$$

Dividing by $B^{19/10}$, and using $2^{19/10} < 4$ and $B \geq 1$, gives $B^{1/10} < 64He^{19h(\beta)/10}$ and therefore

$$\log B < 10\log(64H) + 19h(\beta) < \frac{29}{64}c < c - 4.$$

Here $\log(64H) < c/64$ follows, for example, from $c > (64H)^{20}$ and $H \geq 1$. This contradicts Lucas growth, proving $\beta = n$. Completeness only changes the finite local depths; the real estimate for each normalized detector remains $1 + O(\lambda^{-3})$ for every positive test order. Thus the same simultaneous construction applies.

6.3 Effect on the explicit bounds

These are the two profiles of Theorem 4.6; they alter no retained coordinate or finite algebra degree. The correction-free input map in Section 4 leaves only four finite conic bits. This reduces the finite product's original branch count to 16. The coefficient checker incorporates that exact factor count and the chosen size certificate. The resulting bounds are those in Theorem 4.6.

The finite-anchor source is an exact indexed finite product. Its size exclusion has $2^{2093449}$ factors. The coefficient-height exponent has the form $3\log_2 M$ plus the recorded finite-algebra overhead, which explains the role of shortening this exclusion. The semilocal source instead uses one fixed power of three and has no indexed exclusion. The two profiles therefore serve different degree objectives at their respective fixed numbers of variables.

6.4 Explicit degree and coefficient bounds for the specification

The following bounds concern the expanded polynomial defined by the indexed determinant formula. They do not require expanding its matrices or its final finite product. For a polynomial f, let $\|f\|_1$ denote the sum of the absolute values of its coefficients. Then $\|fg\|_1 \le \|f\|_1\|g\|_1$ and $\|f+g\|_1 \le \|f\|_1 + \|g\|_1$.

Give every base coordinate weight one, and assign a positive integer weight w_i to each algebra generator z_i. Choose the weights so that every nonleading term of its monic relation has weight at most $e_i w_i$. Reduction by these relations never increases weighted degree. Writing $W = \sum_i (e_i - 1)w_i$, the entries of a multiplication matrix M_f therefore have base degree at most $\operatorname{wdeg}(f) + W$: the input basis monomial has weight at most W, and the output basis monomial has nonnegative weight.

For a matrix over the polynomial ring, use the maximum column sum of the coefficient norms of its entries. This norm is submultiplicative. If

$$z_i^{e_i} + \sum_{k<e_i} c_{ik} z_i^k = 0,$$

the block companion description gives

$$\|M_{z_i}\| \le \max\left\{1, \sum_{k<e_i} \|M_{c_{ik}}\|\right\}.$$

Extending to later monic generators preserves this estimate, since multiplication by an earlier element is repeated on diagonal blocks, up to a simultaneous permutation of rows and columns. Thus weights and bounds $\|M_{z_i}\| \le 2^{b_i}$ can be computed recursively from the coefficient DAG. The source compiler records the entire table, and the independent checker verifies the corresponding inequalities.

For the point quartics the weights can be taken to be two. A coefficient bound for the companion matrix is the sum of the bounds for its two nonzero lower coefficients. For each of the subsequent quadratic generators, its nonzero branch has only the constant coefficient LM, so one may take

$$w_i \ge \tfrac{1}{2}\operatorname{wdeg}(LM), \qquad b_i \ge \log_2 \|M_{LM}\|.$$

The zero branch has relation $z_i = 0$ and satisfies the same bounds. Consequently one uniform table applies to all 2^q fixed branches.

Put $n = 2^{22}$ and $m = 13$ for the finite-anchor construction, or $n = 2^{23}$ and $m = 14$ for its semilocal variant, and $T = mn(n+1)/2$ for the largest separating parameter. Suppose the recorded bounds give

$$\operatorname{wdeg}(G) \le w_G, \quad \|M_G\| \le 2^{b_G}, \qquad \operatorname{wdeg}(E_i) \le w_E, \quad \|M_{E_i}\| \le 2^{b_E}.$$

Set

$$b_\lambda = \lceil \log_2(m+1) \rceil + \max_i \{ i \lceil \log_2 T \rceil + b_i \},$$
$$\alpha = w_G + \max_i w_i + W, \qquad \beta = b_G + b_\lambda,$$
$$\alpha_E = w_E + W.$$

These bound the entry degrees and operator norm of $M_{G\lambda_\ell}$, uniformly for all prescribed ℓ and all source branches.

The coefficients of $H(V) = \det(I - VM)$ have base degree at most $A_H = n\alpha$, and its full coefficient norm is at most 2^{B_H} with $B_H = n(\beta + 1)$. Indeed the coefficient norm of a determinant is bounded by the product of its column sums. This bound already includes the sum over determinant permutations; no additional factorial is needed. By multilinearity in columns,

$$R_i(V) = \frac{\partial}{\partial t} \det(I - VM + tM_{E_i}) \Big|_{t=0}$$

has base coefficient degree at most $A_R = (n-1)\alpha + \alpha_E$ and coefficient norm at most 2^{B_R}, where

$$B_R = \lceil \log_2 n \rceil + b_E + (n-1)(\beta + 1).$$

This accounts for the n possible columns containing M_{E_i}.

For each degree case $d \le n$, the Sylvester matrix for $\mathrm{Res}(H_d, H_d')$ has $2d-1$ rows. Every row has coefficient norm at most $d\|H\|_1$. Hence uniform bounds for the reciprocal scale $L = a_d^3 \mathrm{Res}(H_d, H_d')$ are

$$A_L = (2n+2)A_H,$$
$$B_L = (2n+2)B_H + (2n-1)\lceil \log_2 n \rceil.$$

The reciprocal residual has degree at most $n(A_L+1)$ and coefficient norm at most $2^{nB_L + \lceil \log_2(n+1) \rceil}$. The reversed trace residuals have degree at most $A_R + (n-1)(A_L+1)$ and coefficient norm at most $2^{B_R + (n-1)B_L}$. Together with the high-coefficient conditions $a_k = 0$, these give a bound for the sum of squares by doubling the largest degree, doubling the largest logarithmic norm bound, and adding $\lceil \log_2(n+q+1) \rceil$ to the latter. For the reciprocal estimate, no division of coefficient norms is used: its jth coefficient is the polynomial $a_{d-j} a_d^{3j-1} \mathrm{Res}(H_d, H_d')^j$. Its degree is at most jA_L and its logarithmic norm bound is at most $j(3B_H + B_{\mathrm{Res}}) = jB_L$.

The number of final factors is also explicit. Put $n_0 = 4^9$ and $q = 4$ for the finite-anchor source, or $q = 5$ for the semilocal source. For a mask with k nonzero quadratic branches, the rank is $n_k = n_0 2^k$. There are $\binom{q}{k}$ such masks, each with $n_k(1 + mn_k(n_k+1)/2)$ separating-parameter and degree choices. Including the 2^4 original finite parameter choices gives exactly

$$B_{\mathrm{tot}} = 2^4 \left[n_0 3^q + \frac{m}{2} (n_0^3 9^q + n_0^2 5^q) \right]$$

factors. Thus the factor counts are respectively

$$12292021203320302469120, \qquad 119138032421494864216064.$$

Multiplying a uniform branch-degree bound and a uniform logarithmic coefficient-norm bound by this factor count bounds the final product.

For the finite-anchor source the indexed guard is handled before these operations. Its coordinate weight is at most 129, and its coordinate operator logarithmic norm is at most 1048. With $M = 2^{2093449}$, Proposition 5.4 gives

$$\mathrm{wdeg}(\mathcal{E}_{\le M}) \le 2M \cdot 129, \qquad \|M_{\mathcal{E}_{\le M}}\| \le 2^{21M^3}.$$

All these are integer majorants. The emitted power-of-two envelopes for the guard, reciprocal scale, reversed trace and final factor are checked successively by exact integer comparisons. Their final values are

	$\deg P$ upper bound	$H(P)$ upper bound
P_{29}	$2^{2093605}$	$2^{2^{6280501}}$
P_{31}	2^{180}	$2^{2^{184}}$.

The semilocal source records stronger unrounded integer majorants in addition to these convenient envelopes. These estimates concern the expanded polynomial defined by the literal indexed recipe; no expanded matrix or coefficient table is claimed to have been computed.

7 Simultaneous parity and divisibility conditions

Put $K = \mathbb{Q}(\rho)$, where $\rho^2 = 2$, and consider

$$d = 3 - \rho, \qquad D = d^2 = 11 - 6\rho, \qquad E:\ y^2 = x^3 - Dx, \qquad P = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32}\right).$$

We write v_p for the rational valuation with $v_p(p) = 1$. The valuation ν_2 on K above 2 is normalized by $\nu_2(\rho) = 1$, so $\nu_2(2) = 2$. Let $R = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$, and set

$$N(a, b) = a^2 + ab - b^2.$$

Lemma 7.1. *For rational a, b, c, e,*

$$N(a, b) + 6N(c, e) \in R \quad \Longleftrightarrow \quad a, b, c, e \in R.$$

Proof. The binary form N has no nontrivial zero over either $\mathbb{F}_2$ or $\mathbb{F}_3$. Consequently, for $p = 2, 3$ and $(a, b) \neq (0, 0)$,

$$v_p(N(a, b)) = 2\min(v_p(a), v_p(b)).$$

Indeed, divide both inputs by the smaller power of p and reduce modulo p. The two summands $N(a, b)$ and $6N(c, e)$ have valuations of opposite parity when both are nonzero. They therefore cannot cancel. Their sum is p-integral if and only if

$$2\min(v_p(a), v_p(b)) \geq 0, \qquad 1 + 2\min(v_p(c), v_p(e)) \geq 0.$$

Since these minima are integers, these inequalities say that all four inputs are p-integral. If either pair is zero, the same conclusion follows by omitting the corresponding zero term. □

Lemma 7.2. *For $t \in \mathbb{Q}$,*

$$t \in R \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Q}:\ u^2 + 6v^2 - 2w^2 - 3t^2 + 2 = 0.$$

Proof. This is the three-witness semilocal criterion used by the source construction; we recall its arithmetic justification. The Hilbert-symbol formulas show that $(2, 3)_{\mathbb{Q}}$ is ramified precisely at 2 and 3: outside $2, 3$ both parameters are units and the algebra splits; at 3 the symbol is the Legendre symbol $(2/3) = -1$, and reciprocity gives the symbol -1 at 2 since the real place is split. Its quadratic-field embedding criterion [3, Proposition 14.6.7] says that a nonzero rational number Δ is the square of a trace-zero element precisely when Δ is nonsquare in both completions. For a rational square $\Delta \neq 0$, such an element is impossible in a division algebra: $(q - \sqrt{\Delta})(q + \sqrt{\Delta}) = 0$ would make q scalar. For nonsquare Δ, this is the usual local embedding criterion for quadratic fields in quaternion algebras.

For $t \neq 0$, take $\Delta = 1 - 2/(3t^2)$, which is nonzero because $2/3$ is not a rational square. The trace-zero square form is $2x^2 + 3y^2 - 6z^2$. At 2, if $v_2(t) > 0$, then $v_2(\Delta)$ is odd; if $v_2(t) = 0$, then $\Delta \equiv 3 \pmod 8$; and if $v_2(t) < 0$, then $\Delta \in 1 + 8\mathbb{Z}_2$ is a square. At 3, a nonnegative valuation of t gives odd negative valuation of Δ, whereas a negative valuation gives a square in $1 + 3\mathbb{Z}_3$. Hence the embedding criterion holds exactly when $t \in R$. Multiplying $2x^2 + 3y^2 - 6z^2 = \Delta$ by $3t^2$, and putting $v = tx$, $u = 3ty$, $w = 3tz$, gives the displayed equation. The case $t = 0$ has witnesses $(u, v, w) = (0, 0, 1)$. □

Lemma 7.3. *For every nonzero integer n,*

$$\nu_2(x([n]P)) = -6 - 4v_2(n).$$

At the place above 3, every odd multiple $[n]P$ has unit abscissa.

Proof. The displayed coordinates give $\nu_2(x(P)) = -6$, $\nu_2(y(P)) = -9$, and therefore $\nu_2(z(P)) = 3$ for the formal parameter $z = -x/y$. The equation is integral at 2. Its formal logarithm is an isometry on the subgroup of depth strictly greater than $\nu_2(2)/(2-1) = 2$. No minimal-model assertion is needed for this integral formal group [4, Theorem IV.6.4]. Its nonzero logarithm also proves that P has infinite order. Since the logarithm is a homomorphism,

$$\nu_2(z([n]P)) = 3 + 2v_2(n).$$

The Laurent expansion $x = z^{-2}(1 + O(z))$ then proves the formula.

At 3 the equation has good reduction: its discriminant is $64D^3$, a unit. The residue field is $\mathbb{F}_9$, and direct reduction gives $\overline{P} = (\overline{\rho}, 0)$. This point has order two and nonzero abscissa. Every odd multiple therefore has the same reduction and unit abscissa. □

Theorem 7.4. *Let $n_1, \ldots, n_r$ be nonzero integers and write*

$$C = \prod_{i=1}^{r} x([n_i]P) = C_0 + C_1\rho.$$

Then

$$n_1, \ldots, n_r \text{ are all odd} \quad \Longleftrightarrow \quad 2^{3r}C_0,\ 2^{3r}C_1 \in R.$$

Proof. The preceding lemma gives

$$\nu_2(2^{3r}C) = -4\sum_{i=1}^{r} v_2(n_i).$$

Each summand on the right is nonnegative. Thus this expression is nonnegative exactly when every index is odd. The local integer rings at 2 and 3 are respectively $\mathbb{Z}_2[\rho]$ and $\mathbb{Z}_3[\rho]$. At 2 this can also be seen directly from

$$\nu_2(a + b\rho) = \min(2v_2(a), 2v_2(b) + 1),$$

whose two entries have opposite parity. At 3, the integral basis $(1, \rho)$ remains linearly independent modulo 3. Therefore integrality in either local ring is equivalent to integrality of both rational coefficients. If the indices are odd, the preceding lemma makes C a 3-adic unit, so the additional 3-integrality conditions impose no restriction. The converse already follows from the place above 2. □

8 Exact three-adic depth of the anchor

The coefficient size of the physical anchor can be reduced by using its exact local depth. The following calculation concerns only the fixed point and its integral multiples; it is independent of the prime-pattern and global height hypotheses.

Put

$$K = \mathbb{Q}(\sqrt{2}), \qquad D = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and

$$P = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

The prime three is inert in K. Let v be the valuation on $K_3 = \mathbb{Q}_3(\sqrt{2})$ normalized by $v(3) = 1$. As usual, the valuation of zero is $+\infty$.

Proposition 8.1. *For every nonzero integer c, the point $[c]P$ is finite and has nonzero abscissa. Write*

$$f([c]P) = \frac{1}{x([c]P)} = a_c + b_c\sqrt{2}.$$

Then

$$\min\{v_3(a_c), v_3(b_c)\} = \begin{cases} 0, & c \text{ odd}, \\ 2 + 2v_3(c), & c \text{ even}. \end{cases}$$

In the even case the formal parameter $z = -x/y$ satisfies

$$v(z([c]P)) = 1 + v_3(c).$$

Proof. The displayed equation has good reduction at three: its discriminant is $64D^3$, which is a unit. Direct reduction gives

$$\overline{P} = (\sqrt{2}, 0) \in E(\mathbb{F}_9).$$

This is a nonidentity point of order two, with unit abscissa. Hence every odd multiple of P is finite with unit abscissa.

The ordinate of P has valuation one, since

$$y(P) = \frac{3(-220 + 139\sqrt{2})}{32}$$

and its parenthesized factor has nonzero residue in $\mathbb{F}_9$. The numerator of the tangent slope is a unit: modulo three, $3x(P)^2 - D = -D \neq 0$. Thus

$$\lambda = \frac{3x(P)^2 - D}{2y(P)} \quad \text{satisfies} \quad v(\lambda) = -1.$$

The duplication formula now gives

$$\begin{aligned} v(x([2]P)) &= v(\lambda^2 - 2x(P)) = -2, \\ v(y([2]P)) &= v\big(\lambda(x(P) - x([2]P)) - y(P)\big) = -3. \end{aligned}$$

There is no cancellation: the summands in the first difference have valuations $-2, 0$, and those in the second have valuations $-3, 1$. Consequently $v(z([2]P)) = 1$.

For completeness, the needed formal-group valuation is elementary here. The formal group of the integral Weierstrass equation has integral coefficients, and negation sends z to $-z$. Its multiplication series therefore have the form

$$[m](Z) = mZ + Z^3 g_m(Z^2), \qquad g_m(T) \in \mathcal{O}_{K_3}[[T]].$$

If $v(z) \geq 1$ and $3 \nmid m$, the linear term has strictly smaller valuation than every higher term, so $v([m]z) = v(z)$. For $m = 3$, the linear term has valuation $v(z) + 1$, whereas all higher terms have valuation at least $3v(z) > v(z) + 1$. Thus

$$v([3]z) = v(z) + 1.$$

Write $c = 2 \cdot 3^r u$, where $3 \nmid u$. Repeated application to $z([2]P)$ yields $v(z([c]P)) = 1 + r$. In particular, this parameter is nonzero, so $[c]P$ is not the identity. On the formal group, $x = z^{-2}$ times an integral power series with constant term one. It follows that

$$v(f([c]P)) = 2 + 2r.$$

Finally, $1, \sqrt{2}$ form an integral basis for the unramified quadratic extension $K_3/\mathbb{Q}_3$, with linearly independent residues over $\mathbb{F}_3$. For any rational a, b, not both zero, this gives the exact equality

$$v(a + b\sqrt{2}) = \min\{v_3(a), v_3(b)\}.$$

Applying it to the reciprocal abscissas proves both cases. □

Corollary 8.2. *Let $N \geq 1$ and $c \neq 0$. Then*

$$\frac{a_c}{3^N}, \frac{b_c}{3^N} \in \mathbb{Z}_3 \quad \Longleftrightarrow \quad c \text{ is even and } v_3(c) \geq \left\lceil \frac{N}{2} \right\rceil - 1.$$

For a prescribed integer $r \geq 0$, the smallest positive N forcing $v_3(c) \geq r$ is $N = 2r + 1$. At this value the anchor implies

$$|c| \geq 2 \cdot 3^r.$$

Proof. The displayed divisibility condition says that the minimum valuation in Proposition 8.1 is at least N. Since $N > 0$, this excludes the odd case. In the even case it is exactly $2+2v_3(c) \geq N$. The stated rounding follows. When $r \geq 1$, the value $N = 2r$ still admits $c = 2 \cdot 3^{r-1}$, so $2r + 1$ is minimal; for $r = 0$, it is the smallest positive value by definition. The final bound follows from the simultaneous divisibility by two and 3^r. □

9 A scalar normalization without squareclass corrections

Put

$$\beta(X) = 216X^4 + 450459$$

and retain the five-contact polynomial

$$r(T) = T(T - 1)(T - 2401)(32T + 343)(2T - 1).$$

Define

$$r_X = r(\beta(X)), \qquad \delta_X = 1 + (17r_X)^2, \qquad R_X = \frac{102^2 r_X}{\delta_X^2}.$$

The scaling factor is an integer. In particular, this normalization introduces no fixed denominator prime.

Lemma 9.1. *For $X \in \mathbb{Q}$ one has*

$$\beta(X) \in \mathbb{Z} \quad \Longleftrightarrow \quad X \in \mathbb{Z}.$$

For every real X, $\beta(X) \geq 450459 > 2401$, so $r_X > 0$ and $\delta_X > 0$. If $p > 76176$ and $v_p(\beta(X)) < 0$, then

$$v_p(\beta(X)) = 4v_p(X), \qquad v_p(r_X) = 5v_p(\beta(X)), \qquad v_p(R_X) = -3v_p(r_X).$$

In particular, all three valuations are even.

Proof. If X is an integer, so is $\beta(X)$. Conversely, write $X = a/b$ in lowest terms with $b > 0$. Integrality of $\beta(X)$ gives $b^4 \mid 216 = 2^3 \cdot 3^3$. Neither prime exponent admits a nontrivial fourth power, so $b = 1$.

The real inequalities follow from the fourth power and the five positive factors of $r(T)$ when $T > 2401$. At a prime in the stated range, a negative valuation of β forces $v_p(X) < 0$. The unique dominant terms in β and $r(\beta)$ are respectively $216X^4$ and $64\beta^5$. This gives the first two valuation formulas. In this case $v_p(r_X) < 0$, so the dominant term of δ_X is $17^2 r_X^2$, and $v_p(\delta_X) = 2v_p(r_X)$. The final formula follows. The displayed multiple of four is even. □

Lemma 9.2. *If $X \in \mathbb{Z}$, then r_X is a nonzero square in $\mathbb{Q}_2$, $\mathbb{Q}_3$, and $\mathbb{Q}_{17}$. More precisely,*

$$v_2(r_X) = 2, \quad r_X/4 \equiv 1 \pmod 8, \qquad v_3(r_X) = 2, \quad r_X/9 \equiv 1 \pmod 3,$$

and at seventeen either $v_{17}(r_X) = 0$ with square-unit residue, or

$$v_{17}(r_X) = 2, \qquad r_X/17^2 \equiv 9 \pmod{17}.$$

Proof. Since 216 is divisible by eight and by twenty-seven,

$$\beta \equiv 3 \pmod 8, \qquad \beta \equiv 18 \pmod{27}.$$

For the two even factors of $r(\beta)$ one has

$$v_2(\beta - 1) = v_2(\beta - 2401) = 1, \qquad \frac{\beta - 1}{2} - \frac{\beta - 2401}{2} = 1200 \equiv 0 \pmod 8.$$

Their odd parts therefore have square product congruent to one modulo eight. The remaining odd factors have product $3 \cdot 7 \cdot 5 \equiv 1 \pmod 8$. This proves the assertions at two. At three, only the factor β is divisible by three; it has valuation two. Reduction after division by nine gives

$$\frac{r(\beta)}{9} \equiv 2 \cdot 2 \cdot 2 \cdot 1 \cdot 2 \equiv 1 \pmod 3.$$

At seventeen the fourth powers are exactly $0, 1, 4, 13, 16$. Since $216 \equiv 12$ and $450459 \equiv 10 \pmod{17}$, all possibilities are given by the table

X^4	0	1	4	13	16
β	10	5	7	13	15
$r(\beta)$	0	15	2	13	2.

The nonzero residues are squares: $13 = 8^2$, $15 = 7^2$, and $2 = 6^2$ modulo seventeen.

In the remaining case $17 \mid X$. The exact identity

$$32 \cdot 450459 + 343 = 17^2 \cdot 49879, \qquad 49879 \equiv 1 \pmod{17},$$

and $v_{17}(X^4) \geq 4$ show that the factor $32\beta + 343 = 6912X^4 + 17^2 \cdot 49879$ has valuation exactly two and unit residue one. The other four factors are units, with product $10 \cdot 9 \cdot 6 \cdot 2 \equiv 9 \pmod{17}$. This proves the last assertion. The usual local square criteria at two and at odd primes now prove all three square statements. □

Proposition 9.3. *For every integer X, the number R_X is prepared: it is positive and less than one, its reduced denominator is a square, it is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and*

$$v_2(R_X) = 4, \qquad v_3(R_X) = 4, \qquad v_{17}(R_X) \in \{2, 4\}.$$

No squareclass correction is required.

Proof. The contact value r_X is a positive integer. One has $\gcd(r_X, \delta_X) = 1$, and the preceding lemma shows that δ_X is a unit at two and three; it is a unit at seventeen by its definition. Thus $\gcd(102^2 r_X, \delta_X^2) = 1$, and the reduced denominator is exactly δ_X^2.

The quotient $R_X/r_X = (102/\delta_X)^2$ is a rational square. Its local valuations add two at each of $2, 3, 17$, proving both the asserted squareclasses and the exact depths. Finally, since $r_X \geq 1$,

$$0 < R_X = \frac{102^2 r_X}{(1 + 17^2 r_X^2)^2} < \frac{102^2}{17^4 r_X^3} \leq \frac{10404}{83521} < 1.$$

□

9.1 The norm tree and its domain

Fix a positive rational number ρ and six rational tests $t_0, \ldots, t_5$, with their assigned positive integer orders. Start the scalar norm tree with the distinguished node $1 + t_0^2/\rho$. Its other initial nodes are the assigned normalized tests or pair norms. Every later operation is

$$N_\rho(u, v) = u^2 + \rho v^2$$

or the unchanged carrying of an unpaired node. This is the constant-left tree used by the scalar certificate; denote its root by W_ρ.

Lemma 9.4. *Where its rational expressions are defined, $W_\rho > 0$. When all six tests tend to zero, W_ρ tends to one. For fixed nonzero rational ρ, at any finite set of primes the tests can be required sufficiently small that W_ρ is a unit at every prime in that set.*

Proof. The distinguished initial node is positive. At each norm operation its positive descendant contributes a positive square, with a positive coefficient if it is the second argument. Its path therefore remains positive up to the root. When all tests vanish, this path starts at one and all other nodes vanish, so its root is one. For fixed ρ, the tree is a polynomial in the normalized tests with fixed rational coefficients. It is continuous at zero over $\mathbb{R}$ and over every $\mathbb{Q}_p$. Its value at zero is one. The inverse image of $\mathbb{Z}_p^*$ is therefore a neighborhood of zero, proving the final claim. □

There are two valid implementations. One may replace the contact parameter throughout the tree by $\rho = R_X$ and put $b = R_X W_{R_X}^2$. Alternatively, one may retain the original tree with $\rho = r_X$ and normalize only its final pole input:

$$b = R_X W_{r_X}^2.$$

The second choice retains the original test denominators. It also has the quadratic scaling equations

$$\delta_X = 1 + 17^2 r_X^2, \qquad \delta_X V = 102 W_{r_X}, \qquad b = r_X V^2,$$

where the last expression can use the existing square-and-product gates. The first two displayed equations are quadratic. The soundness and completeness proof for this substitution is given below.

For rational inputs, both implementations have $r_X, R_X, \delta_X > 0$ and $W_\rho > 0$. Thus the scalar contact, normalizer, norm-root, and pole-input nonvanishing conditions follow on the stated rational-expression domain. A coordinate used solely to certify one of these nonzero conditions can be removed after its occurrence condition is checked. This observation does not remove unrelated denominator guards in the elliptic-point formulas.

9.2 Soundness and simultaneous completeness

Proposition 9.5. *At every finite scalar contact prime $p > 76176$, the replacement has the same positive odd contact valuation as the original certificate. Consequently it preserves its norm-tree extraction of the assigned contact orders, for either implementation above.*

Proof. At such a contact, $\nu = v_p(r_X) > 0$ is odd. The normalizer satisfies $\delta_X \equiv 1 \pmod{p}$, and 102 is a unit. Hence

$$v_p(R_X) = v_p(r_X) = \nu, \qquad R_X/r_X \in \mathbb{Z}_p^{*2}.$$

For either $\rho = r_X$ or R_X, the two terms in $u^2 + \rho v^2$ have valuations of opposite parity when both are nonzero. There is therefore no cancellation: its valuation is the minimum of $2v_p(u)$ and $\nu + 2v_p(v)$. Every step of the existing constant-left extraction uses exactly this identity, the assigned powers of the contact parameter in the leaves, and the odd-pole exclusion for b. All these valuations are unchanged. Lemma 9.1 also shows that a contact with infinity has even valuation outside the fixed exceptional set, so it creates no omitted odd contact. The inherited height and comparison arguments therefore give $\beta \in \mathbb{Z}$, and Lemma 9.1 gives $X \in \mathbb{Z}$. □

Proposition 9.6. *Under the same arithmetic inputs as the source construction, the scalar block with the input map and normalization above can be satisfied at every integer input, together with the remaining point, height, comparison, and size conditions. The finite squareclass correction parameter can be omitted.*

Proof. Fix an integer input. The numbers r_X, R_X, δ_X are now fixed, before any contact primes are chosen. Proposition 9.3 supplies the sign, local squareclasses, square denominator, and small positive real size formerly obtained by a correction. Include the finite support of the numerator and denominator of R_X, and that of ρ, in the fixed local approximation conditions for the common point H and the scalar point $S - P$.

The required approximation is the same one used in the source. Choose H sufficiently deep at these finite places and in a small real neighborhood of the identity, excluding the identity itself. Shrink the real neighborhood so that its formal parameter is bounded away from zero. Choose $S - P$ still deeper at every fixed finite place and sufficiently close to the real identity. The reciprocal-abscissa tests at these points then tend to zero, and the quotient of their formal parameters tends to zero with H in that smaller neighborhood. Fixed powers of ρ in the denominators do not affect this convergence.

One can impose these conditions with the required congruences on the integer indices: the source first chooses the fixed modulus and depths, then applies its prime-pattern and common-point construction. Increasing the modulus to include the finite support of δ_X is allowed because this support was fixed before that application. It introduces no new family of varying-prime conditions and no stronger prime-pattern theorem. The real compact-group argument permits the same shrinking of the neighborhoods and arbitrarily large scalar index. All other source index and comparison choices are unchanged.

By Lemma 9.4, these choices make W_ρ a unit at every prime in the support of R_X and give $|W_\rho - 1| < 1/2$ at the real place. Therefore

$$0 < b = R_X W_\rho^2 < 9/4 < 17.$$

The factor W_ρ^2 preserves the local squareclasses and, at $2, 3, 17$, preserves the depths of Proposition 9.3. The reduced denominator of b is a square: at a prime in the support of R_X, the root is a unit, while at every other prime its valuation contribution is even. Thus b is prepared. The same inherited rational pole-realization theorem supplies the final pole witnesses. This proves completeness without a correction branch. □

The input polynomial has degree four and can be evaluated with two quadratic gates while preserving the source expression $\beta = 2Y + 3$:

$$Z = X^2, \qquad Y = 108Z^2 + 225228.$$

The gain in any compiled polynomial also depends on its exact occurrence and denominator inventory; the preceding propositions establish the arithmetic substitution independently of that count.

9.3 Number-field scope

All displayed algebraic identities remain valid in characteristic zero. Positivity of the contact and norm root also holds in a number field with a real embedding, after applying that embedding. The local squareclass table, however, concerns rational integer inputs and the three specified completions of $\mathbb{Q}$. It does not establish the corresponding preparation assertions for every algebraic integer in a larger field.

Even the integrality equivalence of Lemma 9.1 does not automatically become an equivalence for rings of integers. Let x_0 be a real fourth root of $1/108$ and $L = \mathbb{Q}(x_0)$. Then

$$\beta(x_0) = 450461 \in \mathbb{Z}.$$

But x_0 is not an algebraic integer, since $x_0^4 = 1/108$. Thus this elementary input map alone does not transfer a definition to $\mathcal{O}_L$. No assertion that the complete rational defining polynomial defines every $\mathcal{O}_L$ is made or follows from this normalization.

10 Exact specifications and reproducibility

The complete coefficient expression graphs and finite algebra relations are supplied in the computation directory. Independent checkers verify the rational source expressions, every monic relation, the guard coverage, the finite alternatives, and the successive degree and coefficient bounds. The producers reproduce the saved specifications exactly. The indexed exclusion products and determinants are specified by finite algorithms; their enormous complete expansions are not needed for either definition or verification of the bounds.

The source arithmetic inputs are separate mathematical premises. The reproduction checks certify the algebraic construction and the stated quantitative implications, not those global arithmetic inputs.

References

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